

Representative Structural Calculation Package

Podium tower setback

Item	Value
Project	Podium tower setback
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	04-podium-tower-setback

Nodes	Members	Loads	Combos	Max disp.	Max util.
84	154	436	28	48.65 mm	3.553

Model And Design Basis

Item	Value
Building type	mixed-use
Description	Six-story mixed-use frame with a broad podium and upper setback tower.
Occupancy	Office
Bounds X/Y/Z	24.00 / 18.00 / 20.40
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	90	0.000 kN	0.000 kN	-9072 kN
L	live	90	0.000 kN	0.000 kN	-4277 kN
WX	wind	64	165.24 kN	0.000 kN	0.000 kN
WY	wind	64	0.000 kN	220.32 kN	0.000 kN
EX	seismic	64	922.85 kN	0.000 kN	0.000 kN
EY	seismic	64	0.000 kN	922.85 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	432.00	1512 kN	777.60 kN	27.54 kN	36.72 kN
2	432.00	1512 kN	777.60 kN	27.54 kN	36.72 kN
3	432.00	1512 kN	777.60 kN	27.54 kN	36.72 kN
4	432.00	1512 kN	777.60 kN	27.54 kN	36.72 kN
5	432.00	1512 kN	777.60 kN	27.54 kN	36.72 kN

6	432.00	1512 kN	388.80 kN	27.54 kN	36.72 kN
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Load Derivation Formula Trace

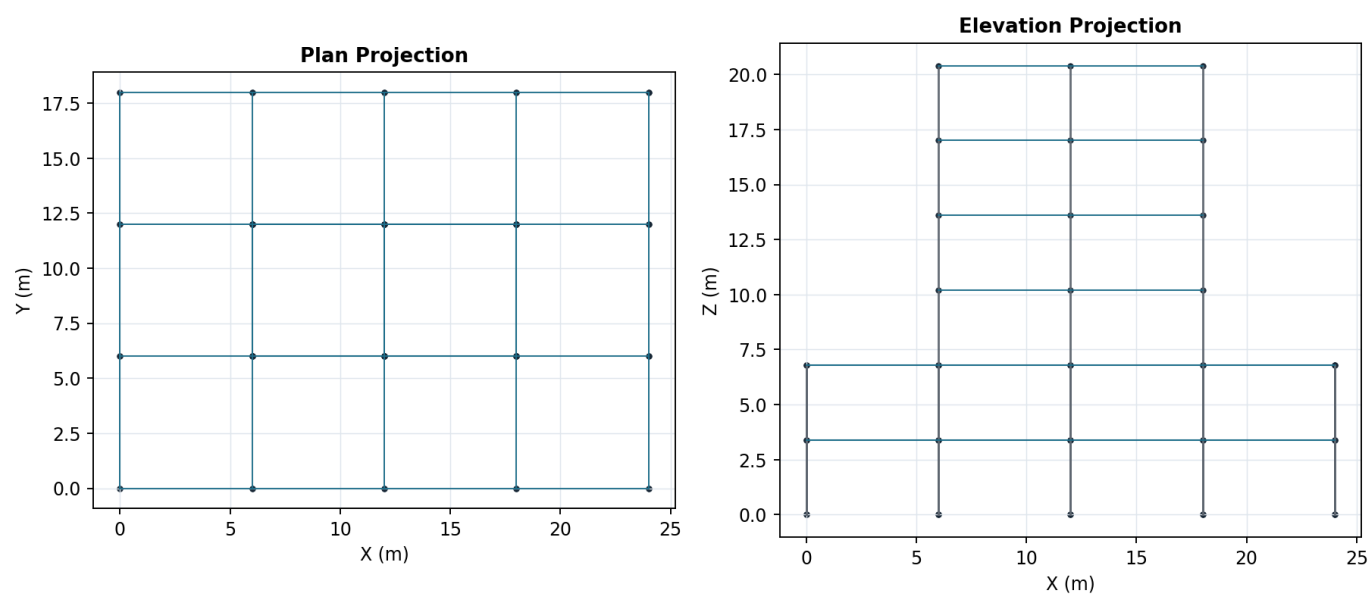
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=office
basis-plan-area	-	max(model width * model depth, 1)	432.00 m2	Bx=24.00 m, By=18.00 m
basis-roof-area	-	roofArea input or typical floor area	432.00 m2	roofArea=432.00 m2
D-ST1	D	$A * q_D$	1512 kN	$A=432.00 \text{ m}^2$, $q_D=3.500 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	777.60 kN	$A=432.00 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	27.54 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=18.00 \text{ m}$, $h=3.400 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	36.72 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=24.00 \text{ m}$, $h=3.400 \text{ m}$
D-ST2	D	$A * q_D$	1512 kN	$A=432.00 \text{ m}^2$, $q_D=3.500 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	777.60 kN	$A=432.00 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	27.54 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=18.00 \text{ m}$, $h=3.400 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	36.72 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=24.00 \text{ m}$, $h=3.400 \text{ m}$
D-ST3	D	$A * q_D$	1512 kN	$A=432.00 \text{ m}^2$, $q_D=3.500 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	777.60 kN	$A=432.00 \text{ m}^2$, $q_L=1.800 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	27.54 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=18.00 \text{ m}$, $h=3.400 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Podium tower setback



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	8.186 mm	0.594	0.000
KDS-ST-02	OK	10.48 mm	0.816	0.000
KDS-ST-04-WX-P	OK	10.68 mm	0.780	0.000
KDS-ST-04-WX-N	OK	10.68 mm	0.780	0.000
KDS-ST-04-WY-P	OK	12.43 mm	0.777	0.000
KDS-ST-04-WY-N	OK	12.43 mm	0.777	0.000
KDS-ST-05-EX-P	OK	42.12 mm	1.295	0.000
KDS-ST-05-EX-N	OK	42.12 mm	1.295	0.000
KDS-ST-05-EY-P	OK	48.65 mm	1.281	0.000
KDS-ST-05-EY-N	OK	48.65 mm	1.281	0.000
KDS-ST-06-WX-P	OK	7.578 mm	0.460	0.000
KDS-ST-06-WX-N	OK	7.578 mm	0.460	0.000
KDS-ST-06-WY-P	OK	9.890 mm	0.457	0.000
KDS-ST-06-WY-N	OK	9.890 mm	0.457	0.000
KDS-ST-07-EX-P	OK	41.45 mm	0.974	0.000
KDS-ST-07-EX-N	OK	41.45 mm	0.974	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	12.64 mm
Max drift ratio	0.00372
Governing	KDS-ST-05-EY-P / story 4

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.400 m	0.035 mm	0.00001	0.002	OK
KDS-ST-01	2	3.400 m	0.066 mm	0.00002	0.004	OK
KDS-ST-01	3	3.400 m	0.023 mm	0.00001	0.001	OK
KDS-ST-01	4	3.400 m	0.034 mm	0.00001	0.002	OK
KDS-ST-01	5	3.400 m	0.037 mm	0.00001	0.002	OK
KDS-ST-01	6	3.400 m	0.113 mm	0.00003	0.007	OK
KDS-ST-02	1	3.400 m	0.048 mm	0.00001	0.003	OK
KDS-ST-02	2	3.400 m	0.092 mm	0.00003	0.005	OK
KDS-ST-02	3	3.400 m	0.032 mm	0.00001	0.002	OK
KDS-ST-02	4	3.400 m	0.046 mm	0.00001	0.003	OK
KDS-ST-02	5	3.400 m	0.033 mm	0.00001	0.002	OK
KDS-ST-02	6	3.400 m	0.130 mm	0.00004	0.008	OK
KDS-ST-04-WX-P	1	3.400 m	0.557 mm	0.00016	0.033	OK
KDS-ST-04-WX-P	2	3.400 m	0.837 mm	0.00025	0.049	OK
KDS-ST-04-WX-P	3	3.400 m	1.378 mm	0.00041	0.081	OK
KDS-ST-04-WX-P	4	3.400 m	1.358 mm	0.00040	0.080	OK
KDS-ST-04-WX-P	5	3.400 m	0.969 mm	0.00029	0.057	OK
KDS-ST-04-WX-P	6	3.400 m	0.644 mm	0.00019	0.038	OK
KDS-ST-04-WX-N	1	3.400 m	0.557 mm	0.00016	0.033	OK
KDS-ST-04-WX-N	2	3.400 m	0.837 mm	0.00025	0.049	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M63	NG	3.553	steel-deflection	KDS-ST-05-EX-P
2	M60	NG	3.553	steel-deflection	KDS-ST-05-EX-P
3	M59	NG	3.532	steel-deflection	KDS-ST-05-EY-N
4	M62	NG	3.532	steel-deflection	KDS-ST-05-EY-P
5	M61	NG	3.532	steel-deflection	KDS-ST-05-EY-N
6	M64	NG	3.532	steel-deflection	KDS-ST-05-EY-P
7	M54	NG	3.101	steel-deflection	KDS-ST-05-EX-N
8	M57	NG	3.101	steel-deflection	KDS-ST-05-EX-N
9	M53	NG	3.076	steel-deflection	KDS-ST-05-EY-N
10	M56	NG	3.076	steel-deflection	KDS-ST-05-EY-P

11	M55	NG	3.076	steel-deflection	KDS-ST-05-EY-N
12	M58	NG	3.076	steel-deflection	KDS-ST-05-EY-P
13	M48	NG	2.380	steel-deflection	KDS-ST-05-EX-N
14	M51	NG	2.380	steel-deflection	KDS-ST-05-EX-N
15	M47	NG	2.342	steel-deflection	KDS-ST-05-EY-N
16	M50	NG	2.342	steel-deflection	KDS-ST-05-EY-P
17	M49	NG	2.342	steel-deflection	KDS-ST-05-EY-N
18	M52	NG	2.342	steel-deflection	KDS-ST-05-EY-P
19	M151	NG	2.027	steel-deflection	KDS-ST-05-EY-N
20	M148	NG	2.021	steel-deflection	KDS-ST-05-EX-P

Member Design Trace Matrix

Item	Value
Checked members	154
Unimplemented members	0
NG / WARN	56/8
Max utilization	3.553

Member	Type	Status	Util.	Formula	Action
M63	steel	NG	3.553	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M60	steel	NG	3.553	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M59	steel	NG	3.532	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M62	steel	NG	3.532	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M61	steel	NG	3.532	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M64	steel	NG	3.532	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M54	steel	NG	3.101	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M57	steel	NG	3.101	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M53	steel	NG	3.076	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M56	steel	NG	3.076	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.

M55	steel	NG	3.076	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M58	steel	NG	3.076	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M48	steel	NG	2.380	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M51	steel	NG	2.380	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M47	steel	NG	2.342	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M50	steel	NG	2.342	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M49	steel	NG	2.342	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M52	steel	NG	2.342	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M151	steel	NG	2.027	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M148	steel	NG	2.021	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	154	64	3.553
Connections	154	88	11.36
Foundations	20	88	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.